MACHINE LEARNING

Supervised learning:

* Naive Baye's: → Algorithm
  ↳ Uses Baye's Theorem.

Baye's Theorem.

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

if

$$P(y/x_1, x_2 \cdots x_n) = \frac{P(x_1, x_2 \cdots / y) \times P(y)}{P(x_1, x_2 \cdots)}$$

Example:

$P(A/B)$: Probability that an email is spam given it contains the word "winner".

→ A : An email is spam.

→ B : Email contains the word "winner".

→ $P(A/B)$: (Posterior) : Probability of A happening, given B is true (Probability it's spam given the word "winner".)

→ $P(B/A)$: (Likelihood) : Prob of happening B given A. (How often "winner" appears in known spam emails).

→ $P(A)$: (Prior) : Probability of A being true or in general.
  ↳ General frequency of spam.

$P(B)$ → (Evidence) : Probability of B being true in general.
  ↳ General frequency of word winner.

CamScanner

* Skitlearn :- library in Python.

↳ Module → Vectorizer: Converts input text into tokens and the makes placement of it.

* Stop words in english:

You are the Winner of the lottery.

→ In above sentence the underlined words are the stop words.

The words left behind are - ["lottery", "Winner"]

Let

Msg 1 = I am the Winner of the lottery.
Msg 2 = I am going to University.
Msg 3 = free lunch Winner today.

→ After removing stop words we left with.

Msg 1 = Winner lottery
Msg 2 = going University
Msg 3 = free lunch Winner today.

Now Place these in to Matrix form.

→ These values are called Vocabulary

	free	Winner	lottery	going	lunch	University	today
Msg 1	0	1	1	0	0	0	0
Msg 2	0	0	0	1	0	1	0
Msg 3	1	1	0	0	1	0	1

The No of Columns shows the size of the vocabulary.

CamScanner

PROBLEM:

R_1 = lunch is ready
R_2 = Winner free lunch.
R_3 = free is good.
} X = train data

R_4 = free. Winner. } y = test-data.

H
S
H
?

Here: size of vocabulary = 6 features = size of vocabulary.

	free	Winner	lunch	ready	good	is
R_1	0	0	1	1	0	0
R_2	1	1	1	0	0	0
R_3	1	0	0	0	1	1

→ Dense Matrix

Now we will find Probabilities.

$P(A) =$

$P(Spam) = 1/3$

$P(Ham) = 2/3$

$P(free/Spam) = \frac{\text{Count of word in Spam} + \alpha}{\text{Total count of words in Spam} + \alpha V}$

$P(free/Spam) = \frac{1+\alpha}{3+\alpha V}$: V = Vocabulary Size = 6, $\alpha = 1$

$= \frac{1+1}{3+1(6)} = \boxed{\frac{2}{9}}$

$P(free/Ham) = \frac{1+\alpha}{6+\alpha \cdot V}$

$= \frac{1+1}{6+1(6)} = \frac{2}{12} = \boxed{1/6}$

$P(free/Spam) = ?$ & $P(free/Ham) = ?$

$P(winner/Spam) = ?$ & $P(winner/Ham) = ?$

Word	Count in Spam	Count in Ham	$P(W/Spam)$	$P(W/Ham)$
free	1	1	$\frac{1+\alpha}{3+\alpha V} = \frac{1+1}{3+6} = \frac{2}{9}$	1/6
winner	1	0	2/9	1/12

$P(winner/Spam) = \frac{1+\alpha}{3+\alpha V} = \frac{2}{9}$

$P(winner/Ham) = \frac{0+\alpha}{6+\alpha V} = \frac{0+1}{6+1(6)} = \boxed{\frac{1}{12}}$

→ α is added in the formula to overcome this 0.

Now We will Calculate the final Score.

Scorce for Spam $= P(\text{Spam}) * P(\text{free}/\text{Spam}) * P(\text{winner}/\text{Spam})$

$= \frac{1}{3} \times \frac{2}{9} \times \frac{2}{9}$

Score for Spam $= 0.0159$

Score for Ham : $P(\text{Ham}) \times P(\text{free}/\text{Ham}) \times P(\text{winner}/\text{Ham})$

$= \frac{2}{3} \times \frac{1}{6} \times \frac{1}{12} = 0.66 \times 0.166 \times 0.0833$

Score for Ham $= 0.0091$

Conclusion:

Score for Spam > Score for ham

Therefore

R_4 = free Winner

will be Spam email.